

Assignment No-04

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Section: 10

Ans to the Q No-1

SR Truth table

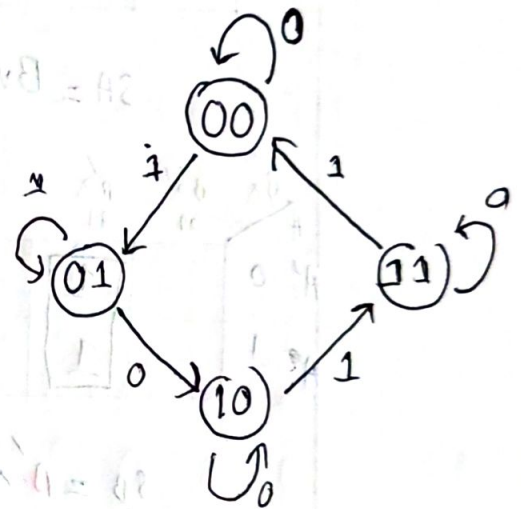
S	R	Q	Q'
0	0	No change	
0	1	0	1
1	0	1	0
1	1	Not possible	

Excitation table

Q	Q'	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Characteristic table

Q	S	R	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	indeterminate
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	indeterminate



Present State		Input		Next State	
A	B	u	v	A ⁺	B ⁺
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	0	1
1	0	0	0	1	0
1	0	1	0	1	1
1	1	0	0	1	1
1	1	1	0	0	0

Flip-flop Inputs			
SA	RA	SB	RB
0	X	0	X
0	X	1	0
1	0	0	1
0	X	X	0
X	0	0	X
X	0	1	0
X	0	X	0
0	1	0	1

Building K map for SA, RA, SB, RB.

A \ Bx	Bx'	B'x	Bx	Bx'
A' 0				1
A 1	X	X		X

$$SA = Bx'$$

A \ Bx	Bx'	B'x	Bx	Bx'
A' 0	X	X	X	
A 1			1	

$$RA = Bx$$

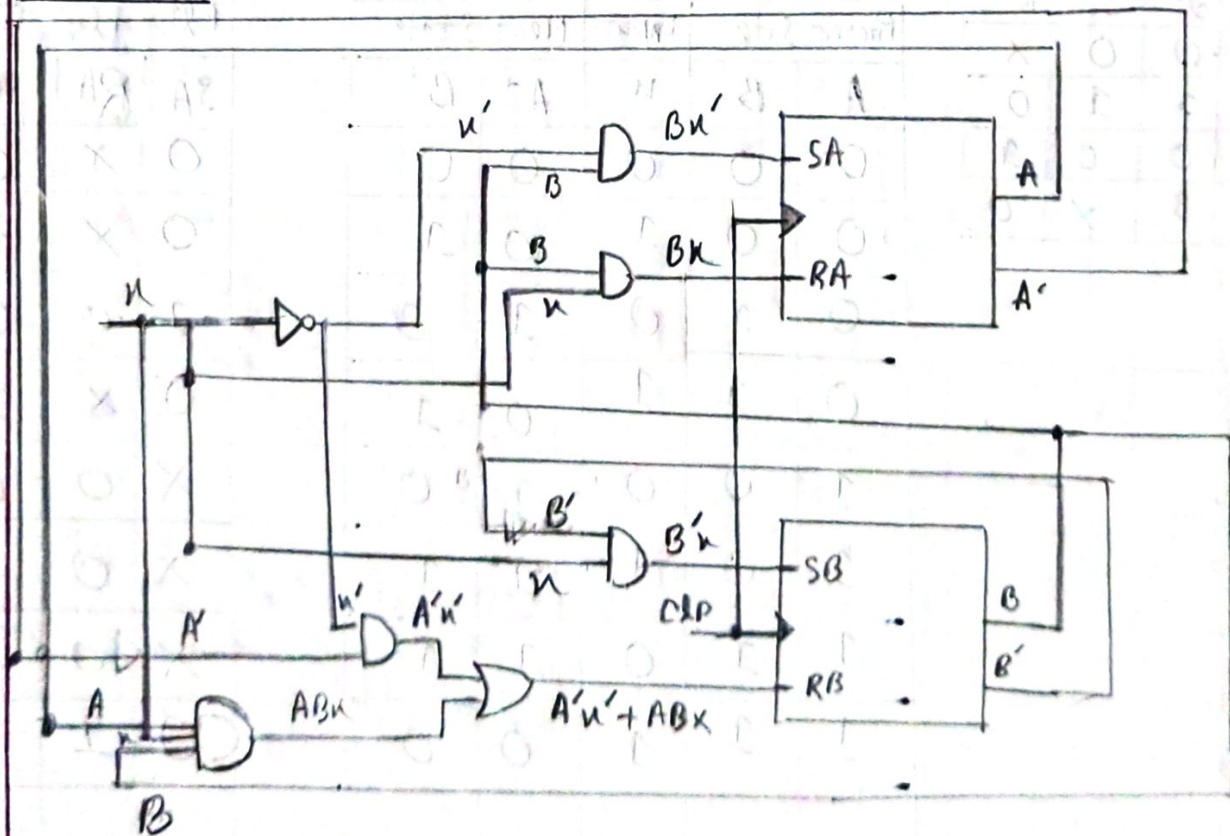
A \ Bx	Bx'	B'x	Bx	Bx'
A' 0		1	X	
A 1		1		X

$$SB = B'x$$

A \ Bx	Bx'	B'x	Bx	Bx'
A' 0	X			1
A 1	X		1	

$$RB = A'x' + ABx$$

Circuit:



Present State A, B, Next state A^+ , B^+ and inputs SA, RA, SB, RB, n.

$$SA = Bn', RA = Bn, SB = B'n, RB = A'n' + ABn$$

A	B	n	SA	RA	SB	RB	A^+	B^+
0	0	0	0	0	0	1	0	0
0	0	1	0	0	1	0	0	1
0	1	0	1	0	0	1	1	0
0	1	1	0	1	0	0	0	1
1	0	0	0	0	0	0	1	0
1	0	1	0	0	1	0	1	1
1	1	0	1	0	0	0	1	1
1	1	1	0	1	0	1	0	0

S-R flip flop Truth table.

S	R	Q	Q'
0	0	No change	
0	1	0	1
1	0	1	0
1	1	Not possible	

State diagram

